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SUPERSEDING
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MILITARY SPECIFICATION

PASSAGEWAY, INSULATED, MEDICAL UNIT,

SELF CONTAINED TRANSPORTABLE (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an insulated passageway fabricated from coated cloth and insulating foam and supported by an aluminum frame assembly. The passageway has the following dimensions: width 5 feet (1.5 m); height, 7 feet (2.1 m); length, 14 feet (4.3 m).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Steel, and Seals
DDD-L-20	-	Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	-	Strapping, Nonmetallic (and Connectors)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic
MIL-C-43285 - Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated)
MIL-R-44024 - Repair Kit, Inflatable Shelter, (MUST)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking for US Military Property
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-4-1694 - Adhesives
5-4-3578 - Passageway, Insulated, 14 Feet (MUST) Drawing List

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3574 - Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Alternate components. Components shall be as specified on the applicable drawings and as specified herein. For components for which it is stated "or equal", if the contractor proposes to use an item considered to be equal to the component specified, prior to its use the contractor shall furnish a sample of the component, with supporting data to the contracting officer for subsequent evaluation by the responsible military agency. The supporting data required shall prove the functional equivalence and design compatibility of the item proposed to be used. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article of production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample of the complete passageway to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.3 Initial production testing sample. When specified (see 6.2), the contractor shall furnish a sample for initial production inspection and approval (see 4.4 and 6.4).

3.4 Inspection comparison testing sample. When specified (see 6.2), the contractor shall furnish a sample for comparison inspection and approval (see 4.5 and 6.5).

* 3.5 Materials and components. Materials and components shall conform to the specifications, standards and drawings listed in section 2 and as specified herein (see 6.8).

* 3.5.1 Cloth, polyester, coated. The coated polyester cloth shall be as specified on the applicable drawing, with the following exceptions:

a. That the adhesive used in preparing only the adherability test specimens requirements of MIL-C-43285, shall be as specified in 3.5.5.

b. That the type I, class 1 coated fabric shall also meet the 200°F (93°C) adherability requirement of MIL-C-43285 with not more than 1/16 inch (2 mm) slippage.

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3.5.2 Thread, polyester. Dyed thread shall show fastness to weathering equal to or better than the standard shade sample. When no standard shade sample is available, the dyed thread shall show good fastness to weathering.

3.5.3 Overcoating materials. Materials used for overcoating (see 3.6.8), shall be air dry coatings meeting color and compositional requirements of the green top coating and pale green top coating of MIL-C-43285. When applied to a sample of coated cloth, as used in the passageway, and dried, the resulting sample shall meet the requirements of MIL-C-43285 for blocking resistance and flame resistance.

* 3.5.4 Dusting compound. The dusting compound shall be whiting, talc, or other finely divided mineral material. The compound shall not support mildew growth. The contractor shall submit a certificate of compliance to the contracting officer that the dusting compound conforms to these requirements (see 4.6.1.3).

* 3.5.5 Adhesive. The adhesive shall be as specified on Drawing 5-4-1694. Containers of the adhesive base and the curing agent shall be marked with the manufacturer's batch number and an expiration date and shall be stored in an area where the temperature shall not exceed 70°F (21°C). The batch number identifies that quantity of material manufactured at the same time using the same raw materials or subjected to some unit chemical or physical mixing process intended to make the final product homogeneous. The expiration date, date after which the adhesive or curing agent shall not be used, shall be three months after the date of manufacture. In addition, containers of adhesives and curing agent received in one shipment, shall be identified and segregated to insure their use prior to later deliveries. Before use, each lot of adhesive and curing agent shall be tested for dead load shear strength and dead load peel strength. The adhesive mix shall brush out smoothly without producing drag or showing stringiness. An adhesive lot is approved for use during the 14 days following the date of acceptance (see 4.6.1.1). At the end of 14 day period, unused adhesive may be retested as long as the expiration date of the adhesive and the curing agent has not been reached.

* 3.5.5.1 Dead load shear strength. There shall be not more than 1/16 inch (2 mm) slippage at 200°F (93°C) when the adhesive is tested as specified in 4.7.1.1. This requirement is for all adhesives.

* 3.5.5.2 Dead load peel strength. There shall be not more than 1/2 inch (13 mm) peel at room temperature when the adhesive is tested as specified in 4.7.1.2. This requirement is only for strapping, patching and splicing adhesive.

3.5.6 Polyurethane foam. The polyurethane foam shall be a flexible and flame-retardant foam conforming to the following requirements when tested as specified in 4.6.1.2:

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Density, lb/cu ft (kg/m^3)	2.6 maximum (41.6)
Indentation load, lb (N), delfection at 25%	25 ± 5 (111 ± 22)
50% compression set	20 percent maximum
Tear resistance, lb - force per lineal inch (N/m)	2 minimum (350)
After-flame time, seconds	
Original	2 maximum
After oven aging	2 maximum
After steam autoclave	2 maximum

* 3.6 Design and construction. Design and construction of assemblies and components shall conform to the drawings listed on Drawing 5-4-3578 and as specified herein.

3.6.1 Coated cloth components. All visible outside surfaces of coated cloth components and assemblies shall be olive green and all visible inside surfaces shall be pale green.

3.6.2 Stitching machine. All stitching shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch (2 to 3 stitches/cm).

3.6.2.1 Thread breaks. Thread breaks shall be overstitched not less than 1 inch (25 mm) at each break. Any thread break noted during inspection shall be repaired by overstitching the existing stitching, starting from a distance of 1 inch (25 mm) before the break, across the defective area, to a distance of 1 inch (25 mm) beyond the break. The ends of repair stitching are not required to be backstitched.

* 3.6.2.2 Stitching ends. The ends of stitching shall be overstitched not less than 1 inch (25 mm) except where ends are turned under in a hem, held down by other stitching or cementing. Where stitching is performed automatically, stitch patterns such as box and box with cross-stitch, 3 or more tying, overlappin or backstitches shall be used to secure the ends.

* 3.6.2.3 Repair of skipped stitches or run-offs. Two or more consecutively skipped stitches or run-off stitches shall be repaired by overstitching. Overstitching shall be accomplished for a distance of not less than 1 inch (25 mm) before, to a minimum of 1 inch (25 mm) beyond, the affected area of skipped stitches or run-offs.

3.6.2.4 Lubrication of thread. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited. Water or air spraying for needle cooling is permissable.

* 3.6.3 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the materials.

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3.6.4 Setting of rivets. Rivets shall be securely set and neatly headed.

3.6.5 Setting of snap fasteners. If a hole is punched before inserting the male or female part of the fastener, it shall be smaller than the outside diameter of the fastener tube so that the tube must be forced through the hole. The fasteners shall be securely clinched.

3.6.6 Fusing of webbing ends. Cut ends of nylon webbing shall be fused and, after fusing, shall be free from frayed edges and loose fibers.

* 3.6.7 Fusing of polyester and nylon cord. The cut ends of braided cords shall be fused to prevent fraying.

* 3.6.8 Cementing. Cementing of assemblies shall be in accordance with the following procedure:

a. Surface preparation. Surfaces to be bonded shall be clean and dry prior to application of the adhesive. Coated cloth surfaces to be bonded shall be thoroughly cleaned by rubbing with toluene-wetted clean cloths or pads, which may contain a mild, bonded abrasive of 80, or finer, grit size or may be of a coarse weave (as for scouring). If abrading is exercised, there shall be no scuffing or abrading of the polyester base cloth. Separate clean cloths and clean solvent shall be used for final cleaning wipes. Abrasive coated cloths or pads shall not be used directly on stitching at seams. Foam surfaces to be bonded to coated fabric need not be cleaned unless soiled or dust ladened. Uncoated sides of fabric being bonded (bladders, inner casing patches, etc.) shall be toluene-cleaned, using a clean cloth and clean toluene and allowing the surfaces to dry.

b. Adhesive mixing. The ratio of the adhesive mixture shall be as specified on the drawings. Prior to extracting quantities of the base and accelerator to be mixed, any settlement shall be uniformly redispursed. Containers for mixing, transfer and use shall be clean prior to each filling. Adhesive in the containers shall be covered during transit to work stations and during sustained periods of non-use to prevent loss of solvent by evaporation. After mixing, the adhesive shall be used only while sufficiently spreadable to obtain an even coat. A mix tending to produce drag or showing stringiness when being brushed out has started to cure and must be discarded.

c. Adhesive application. Except for coated cloth-foam laminates, a minimum of two coats of mixed adhesive shall be evenly spread on the cleaned surfaces to be bonded. The first coat shall be dry to the touch before the application of the second coat. Prior to assembly, the adhesive coated surfaces shall not be allowed to be contaminated by a visible layer of dust or other foreign matter. Applied adhesive shall be free of heavy smears or

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excessively thick lap edges (resulting from insufficient spreading), thin starved areas, uncoated areas or other defects characteristic of improper application. The second coat shall be dried as appropriate for the assembly method to be used; i.e., to a tacky (not wet) aggressive grab condition for the two-way, dry stick method; or to a low-tack (dry condition) for two-way stick, solvent reactivation method. Adhesive application on coated cloth-foam laminates shall be either in one coat application to each surface followed by two-way, dry stick assembly, or in one heavy coat application followed by wet-tack assembly.

d. Assembly. Assembly of components shall be made when the tack and grab of the adhesive coated surfaces are in a suitable range and a condition of either excess solvent in the adhesive layers or excessive drying or open time shall be avoided. Surfaces shall be joined in the locations specified on the drawings, applying adequate pressure and rolling or barring firmly to produce an adequate bond. Care shall be exercised to avoid finger contact in the adhesive coated areas along the edges of seams, patches, or other bonded parts. Precaution should be taken to prevent adhesive application, assembling or curing of an assemb at a temperature below 60°F (16°C). The assembled work shall not be moved too soo or in such a way as to result in edge lifting or a decreasing strength of bond until the adhesive has set. Surfaces shall be assembled on the same day that they were coated, or a third coat of adhesive shall be applied to each surface. When surfaces have been assembled in an incorrect location, the adhesive bond may be immediately separated and reset at the proper location. To facilitate separation, toluene may be applied at the bond line by squirt can or atomizer and the component then peeled away as the adhesive softens. The excess toluene solvent shall be allowed to evaporate before resetting the surfaces. When incorrectly located parts are separated and the adhesive is not in an active condition or has transferred to the other surface, the affected surface shall be given an additional coat of adhesive before reassembling. Any adhesive residue shall not extend further than 1/2 inch (13 mm) beyond the edges of seams, strapping, splices, patches or other bonded parts. Any exposed adhesive within the 1/2 inch (13 mm) allowance shall be overcoated with material specified in 3.5.3. Overcoated areas shall be dusted with the dusting compound specified in 3.5.4. There shall be no unsightly smearing or spreading of adhesive or overcoating and care shall be exercised to insure that surrounding areas are kept clean.

3.6.9 Splicing of cloth. Splicing of cloth panels shall be allowed only as specified herein.

3.6.9.1 End panel and side panel assemblies. A maximum of one splice per interior and exterior cloth panel of each end panel assembly shall be allowed. A maximum of two splices per interior and exterior cloth panel of each side panel assembly shall be allowed. The shortest length of any spliced piece shall be not less than 3 feet (1 m) and pieces shall be joined as specified for panel lap seams shown on applicable drawings.

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3.6.9.2 Floor assembly. A maximum of two splices per floor assembly shall be allowed. The shortest length of any spliced piece shall be not less than 3 feet (1 m) and pieces shall be joined as specified for panel lap seams shown on the applicable drawing.

3.6.10 Restoration of damaged components. Components damaged during fabricatior when adjudged repairable and when approved by the contracting officer, shall be repaired in accordance with requirements specified herein. Cementing shall be performed as specified in 3.6.8. All patches shall have rounded corners.

3.6.10.1 Interior and exterior cloth panels. Cuts, holes or tears to cloth panels shall be repaired by applying a patch extruding not less than two inches beyond the damaged area using the same material as specified for the damaged assembly. A maximum length of 3-1/2 inches (89 mm) of damage shall be allowed and not more than two patches with a maximum size of 36 square inches (232 sq cm) per patch shall be allowed per assembled inner and outer cloth panels.

3.6.10.2 Fabric to foam laminations. Cuts, tears and holes shall be repaired by cementing a patch of the same materials as the damaged laminate and the inner and outer cloth shall extend not less than 2 inches (51 mm) beyond the damaged area. Not more than two patches with a maximum size of 24 square inches (155 sq cm) per patch shall be allowed per individual passageway or floor assembly.

3.6.11 Welding. All welding shall be as specified on the drawings. Materials to be welded shall be clean and free from scale, corrosion, oil, water and other foreign matter. Welds shall be sound, smooth, free from flash and show good fusion.

* 3.7 Finish of metal components. All metal components shall be finished as specified on the drawings. Finish painted coats shall level out to an adherent, continuous and uniform film without runs, wrinkles, streaks or areas of no film. Any coating damaged during assembly or examination shall be touched up by priming and painting as on prescribed drawings.

3.8 Marking.

3.8.1 Nomenclature. The application of the letters "US" and the nomenclature marking on the passageway shall conform to type IV, class 9 of DDD-L-20, in the size characters and locations shown on the drawing. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.8.2 Identification labels. Identification labels shall conform to type II, class 6 of DDD-L-20 and shall be 3-1/2 inches (89 mm) by 5-1/2 inches (140 mm) minimum with a blank space of at least 1-1/4 inches (32 mm) provided on the bottom of the label for the Department of Defense inspector's stamp. Identification labels shall be stitched to the passageway in the location shown on the drawing.

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3.8.3 Identification marking. Frame component assemblies (support, arch, and beam) shall be marked for identification in accordance with MIL-STD-130.

3.9 Workmanship. Cloth components shall be free of holes, cuts, or tears, thin or weak places caused by abrasion, exposed fabric, blisters, tunnels, unadhered pockets, loose edges, or delamination of coating, creases, wrinkles or coating missing. Webbing shall contain no frayed or scalloped edges. Rope components shall be free of cuts, abrasions or breaks. Thread tension shall be maintained so that there will be no loose stitching and seam allowances shall be maintained with seams properly sewed so that no run-offs, pleats or open seams shall result. Care shall be taken in sewing so that there shall be no needle chews. All thread ends shall be trimmed to 1/2 inch (13 mm) or less. Metal components shall be free of burrs, sharp edges, corroded areas and shall not be broken or malformed.

* 3.10 Repair kits with storage box. When required, repair kits with storage box shall conform to MIL-R-44024 (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Quality conformance inspection (see 4.6).

4.3 First article inspection. When a first article is required, the complete passageway assembly shall be examined for defects listed in tables I through III and dimensions specified. The presence of any visual defect or any dimension not within the specified requirement shall be cause for rejection of the first article.

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4.4 Initial production sample inspection. When an initial production sample is required, the complete passageway assembly shall be examined for defects as specified in tables I through III and dimensions specified. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the initial production sample.

4.5 Comparison sample inspection. When a comparison sample is required, the complete passageway assembly shall be examined for defects as specified in tables I through III and dimensions specified. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the comparison sample.

4.6 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.6.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.6.1.1 Adhesive testing. Each lot of adhesive shall be tested as specified in 4.7.1.1 for dead load shear strength, and as specified in 4.7.1.2 for dead load peel strength. The inspection lot of adhesive shall consist of all containers of adhesive base from the same batch, and all containers of accelerator from the same batch. The quantities of base and accelerator making up the lot shall be deemed appropriate for production use within the prescribed 14 days usage limitation as specified in 3.5.5. The sample unit for the dead load shear test shall be one cloth lap seam and two determinations shall be made per sample lap seam. The sample unit for the dead load peel test shall be one cloth laminate and one determination shall be made per sample laminate. Any nonconformance to the requirements of 3.5.5.1 or 3.5.5.2 shall be cause for rejection of the representative lot of adhesive. Containers of the adhesive base and the accelerator for each acceptable lot of adhesive shall be marked with the 14 day usage date and with a lot identity.

4.6.1.2 Polyurethane foam.

* 4.6.1.2.1 Foam testing. The polyurethane foam shall be tested as follows for compliance with the requirements of 3.5.6. Any nonconformance shall be cause for rejection of the lot of material.

Characteristic	Test method
Density	ASTM D 3574, sec 6-11
Indentation load, deflection	ASTM D 3574, sec 12-18
50% compression set	ASTM D 3574, sec 25-30
Tear resistance	ASTM D 3574, sec 45-51
After flame time	
Original	FED-STD-191 Method 5905
After oven aging, 22 hours at 284°F (140°C)	ASTM D 3574, sec 88-92
After steam autoclave, 5 hours at 250°F (121°C)	ASTM D 3574, sec 83-87

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4.6.1.2.2 Lot. For purposes of sampling, a lot shall consist of all material produced under like conditions in one unchanged process by one manufacturer and submitted for acceptance at one time.

4.6.1.2.3 Sampling for testing. A sample shall be selected as listed below and subjected to the tests described in 4.6.1.2.1.

a. Rolls - A sample roll shall be selected at random from the lot. Specimens of sufficient size to conduct the tests shall be taken from the sample after removal of two thicknesses of material from the roll.

b. Sheets - Three bundles shall be selected, the top three sheets of each bundle removed, and a sample taken at random from the remainder of each bundle.

4.6.1.3 Dusting compound certification. The certificate of compliance submitted as proof of compliance with the dusting compound requirements in 3.5.4 shall be examined and validated.

4.6.2 In-process inspection. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

a. That the color of all visible exterior and interior surfaces of coated cloth components and assemblies are in compliance with 3.6.1.

b. That no lubrication is applied to the polyester sewing thread prior to or during sewing operations for compliance with 3.6.2.4.

c. That holes punched to receive snap fasteners are in compliance with 3.6.5.

d. That the use of adhesive is in compliance with cementing instructions specified in 3.6.8.

e. That welds and welding are in compliance with 3.6.11 and applicable drawings.

f. That metal components are finished as specified in 3.7 and applicable drawings.

4.6.3 End item inspection. The inspection lot shall consist of all passageways, or all complete frame assemblies, as applicable, offered for inspection at one time. The sample unit shall consist of a passageway, or complete frame assembly, as applicable.

4.6.3.1 Visual examination of passageways. Visual examination of the passageway shall be made for defects in table I. The inspection level shall be II, and the acceptable quality levels (AQLs), expressed in terms of defects per hundred units, shall be 15 for major defects and 100 for total defects (major and minor combined).

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TABLE I. Classification of defects for visual characteristics for the passageway

Examine	Defect	Classification Major	Minor
Cloth, coated	Hole, cut, or tear; exposed fabric; delamination of coating; crease or wrinkle; blister; tunnel; scratch or abrasion	X	
	Lump or heavy coated area		X
Webbing	Frayed or scalloped edge		X
	End not fused		X
Rope, polyester	Cut, or broken or abraded yarn	X	
	End not fused		X
Hardware	Broken	X	
	Malformed	X	
	Corroded area		X
	Burr or sharp edge which may cause injury in handling or damage to fabric	X	
Snap fasteners	Fastener fail to snap in a closed position or to remain firmly closed	X	
Cementing	Pucker; loose edge, unadhered pocket or tunnel		X
	Excessive adhesive in sags or lumps evident through outer layer of fabric		X
	Adhesive exposed beyond the over-coated area:		
	- on exterior surface	X	
	- on interior surface		X
Becket loops	Incorrect number, wrong length, wrong material, not attached as specified	X	
Touch and close fastener tapes	Hook or pile incorrectly positioned, damaged, wrong width, not attached as specified	X	

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TABLE I. Classification of defects for visual characteristics for the passageway (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitching	Open seams:		
	- 1-inch (25 mm) or more	X	
	- for less than 1-inch (25 mm)		X
	NOTE: A seam shall be classified as open when or more skipped or run-off stitches occur. On double stitched seams, a seam shall be considered open when either one or both sides of a seam are open.		
	Thread breaks overstitched less than 1 inch (25 mm) before and beyond break		X
	Needle chew resulting in cut, tear or hole	X	
	One or more required rows of stitching omitted (except on box stitching)	X	
	Skipped stitches or run-offs overstitched less than 1 inch (25 mm)		X
	NOTE: Skipped stitches and run-offs not overstitched shall be classified as open seams		
Seam type	Wrong seam type	X	
Stitch type	Wrong stitch type	X	
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches (152 mm)		X
	Tight as evidenced by puckering on fabric for more than 6 inches (152 mm)	X	
Stitches per inch	Two or more stitches less than specified		X
	More than maximum specified:		
	- one or more stitches		X

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TABLE I. Classification of defects for visual characteristics for the passageway (cont'd)

Examine	Defect	Classification Major	Classification Minor
Stitching margin	Larger than specified by more than 3/16 inch (5 mm):		
	- for more than 2 inches (51 mm), but not more than 4 inches (102 mm) in length		X
	- for more than 4 inches (102 mm) in length	X	
Stitching gage	Not as specified		X
Stitching ends	Not overstitched	X	
	Overstitched less than 1 inch (25 mm) (except where ends are held down by other stitching, cementing or turned under in a hem)		X
Boxstitching	Wrong stitch type	X	
	Incomplete:		
	- one row or stitching omitted		X
	- two or more rows of stitching omitted	X	
Components and assembly	Any component part not fabricated of applicable referenced material	X	
	Any component or assembly with repairs beyond specification allowances	X	
	Any component that will not assemble with its mating component	X	
Marking; labels	Omitted, incorrect, illegible, or misplaced		X

* 4.6.3.2 Dimensional examination. Dimensional examination of passageways shall be made for defects in table II. The inspection level shall be S-3 and the AQLs, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 40 for total defects (major and minor combined).

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TABLE II. Classification of defects for dimensional characteristics

Examine	Defect	Classification Major	Classification Minor
Overall dimensions (unless otherwise specified)	Smaller than specified dimensions less applicable minus tolerance indicated on drawings	X	
	Larger than specified dimensions and applicable plus tolerance indicated on drawings		X
Location dimensions (unless otherwise specified)	Not within specified tolerance		X

4.6.3.3 Visual examination of frame assemblies. Visual examination of frame assemblies shall be made for defects in table III. The inspection level shall be II and the AQLs, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for major and minor defects combined.

TABLE III. Classification of defects for visual characteristics for frame assemblies

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished	X	
	Not adherent; e.g., blistered, peeled or flaky		X
	Coating not continuous or uniform; i.e., not coated or area of no film		X
	Corroded area		X
	Scratch, dig or abrasion exposing bare metal		X
Construction and workmanship	Component missing	X	
	Any component not fabricated of applicable referenced material	X	
	Split, puncture, burr, rough or sharp edge, deformed	X	
Welding	Missing, incomplete, burnt through, cracked, porous, fractured or otherwise not fused	X	
	Not type specified, not smooth, flux and scale not removed		X

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TABLE III. Classification of defects for visual characteristics for frame assemblies (cont'd)

Examine	Defect	Classification Major	Minor
Rivets and riveting	Rivet missing	X	
	Not specified type, size		X
	Rivet not peened; i.e., can be removed by hand	X	
Marking	Missing, incomplete, illegible, or incorrect		X

4.6.3.4 Dimensional examination of frame assemblies. Dimensional examination of complete frame assemblies shall be performed. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.6.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with section 5 requirements. Defects shall be scored in accordance with table IV. The sample unit shall be one shipping container fully packaged, except that it need not be closed. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE IV. Packaging inspection

Examine	Defect
Marking	Omitted, incorrect or illegible Not in accordance with the specification
Materials	Any component missing; any component damaged
Workmanship	Bow sections not secured; guy lines not individually coiled and secured against unwinding

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4.7 Methods of inspection.

4.7.1 Adhesive testing.

4.7.1.1 Adhesive dead load shear strength.

4.7.1.1.1 Preparation of sample lap seam. Two rectangular swatches of type I, class 1, MIL-C-43285 coated fabric shall be cut, each approximately 8 inches by 10 inches (203 by 254 mm) with warp threads running parallel to the 10 inch (254 mm dimension. The swatches shall be cut from areas of the coated fabric that are free of waviness or distortion of the weave. A quantity of the adhesive base and the accelerator, sufficient to fabricate the test samples, shall be selected from the adhesive lot. The two swatches shall be joined with the adhesive to form a lap seam wherein the chlorosulphonated polyethylene side of one swatch is adhered to the chloroprene side of the other. The seam shall be made by overlapping the 8 inch (203 mm) ends for a distance of 2 inches (51 mm), with care exercised that the warp directions of the two swatches are aligned. The surfaces to be bonded shall be cleaned, adhesive shall be applied, and the assembly shall be made in the same manner used in production (see 3.6.8). Allow the seam to cure at 70 to 75°F (21 to 24°C) for 7 days prior to testing. Adhesives may be tested in less than 7 days curing, and if found to meet the requirements, will be acceptable.

4.7.1.1.2 Specimens. One-inch (25 mm) wide strips shall be cut from the sample lap seam prepared as specified in 4.7.1.1.1, to provide 1-inch (25 mm) wide by 18-inch (457 mm) long specimens with warp threads running parallel to the 18-inch (457 mm) length.

4.7.1.1.3 Procedure. One end of each specimen shall be anchored in a suitable grip. A load of 40 pounds (18 kg) shall be fastened to the other end of the specimen. The anchoring grip and the load application grip shall be of a type which precludes failure of the specimen in or near the grips. The grips shall not be applied any closer than 2 inches (51 mm) to the lap seam and shall hold the specimen uniformly across the 1-inch (25 mm) dimension without twist. A suitable gage mark (e.g., ball point line) shall be made at the lap seam edge prior to loading to assist in determining the amount, if any, of relative seam slippage. The two specimens shall be tested for 4 hours at 200°F (93°C). After removal of load, a second gage mark shall be made at the lap seam edge. The amount of slippage shall be reported as the distance measured to the nearest 1/16 inch (2 mm) between center lines of the gage marks for compliance with 3.5.5.1. Slippage greater than 1/16 inch (2 mm) shall constitute a failure.

4.7.1.2 Adhesive dead load peel strength.

4.7.1.2.1 Preparation of sample laminate. A sample laminate shall be prepared in the same way as described in 4.7.1.1.1 except that (a) the two pieces of cloth shall have adhesive applied over most of the surfaces of both pieces except for approximately 2 inches (51 mm) at one end and (b) the two pieces shall be assembled in a full overlap in lieu of the 2 inch (51 mm) lap.

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4.7.1.2.2 Specimens. One-inch (25 mm) wide strips shall be cut from the laminate. The uncemented end will provide tabs for conduct of the peel test. The tabs shall be further separated by hand for at least 1 inch (25 mm) prior to test.

4.7.1.2.3 Procedure. The specimen shall be subjected to a test load of 1-pound (0.45 kg) (including weight of grip) in dead load peel for 10 minutes at a temperature of 70 to 75°F (21 to 24°C). The anchoring grip and force application grip shall hold the tab ends uniformly across the 1-inch (25 mm) dimension and the force shall be applied uniformly without twist. Before application of the load, a suitable gage mark shall be made at the line of peel on one of the tabs. At the end of 10 minutes of test, the force shall be removed and a second gage mark made in the same way on the same tab. The distance between the marks shall be measured to the nearest 1/16 inch (2 mm) as the amount of peel for compliance with 3.5.5.2. Peel greater than 1/2 inch (13 mm) shall constitute a failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The complete passageway assembly shall be disassembled. The frame components shall be neatly bunched and secured with flat strapping or tape and guy lines shall be individually coiled and secured against unwinding. The fabric body shall be neatly and compactly folded.

* 5.1.2 Commercial. Complete passageway assemblies shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each passageway assembly, preserved as specified in 5.1 shall be packed in a snug-fitting, skidded, cleated plywood shipping container conforming to overseas type, type 2 load of PPP-B-601. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Each passageway assembly, preserved as specified in 5.1, shall be packed in a snug-fitting cleated plywood shipping container conforming to domestic type, style A or B of PPP-B-601, or class 2, style G of PPP-B-640. Boxes conforming to PPP-B-601 shall be provided with skids and boxes conforming to PPP-B-640 shall be provided with a base. The base shall be made from minimum of three longitudinal (full length of the bottom of the box) 1 by 6 inch (25 by 152 mm) lumber secured to three (full width of the box) 2 by 4 inch (51 by 102 mm) skids placed one, 1/6 the distance from each end and the third equidistant between the end skids. The 1 by 6's (25 by 152 mm) shall be secured to the 2 by 4's (51 by 102 mm) with at least 3 spaced 8D box or common nails and the nails clinched. The box shall be secured to the base with flat head asphalt roofing nails provided

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with flat metal washers. The box and cover shall be secured to the base with not less than three girth straps. The strapping shall be 3/4 by 0.020 inch (19 by 5 mm) steel strapping conforming to QQ-S-781 or minimum 850 lb. (386 kg) breaking strength strapping conforming to type II or III of PPP-S-760. Edge protectors shall be placed between the edge of the box and steel straps and buckles shall not be used for nonmetallic straps.

* 5.2.3 Commercial packing. Passageway assemblies, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

* 5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The insulated passageway is intended to permit passage between components of the MUST System without exposure to the elements. The passageway provides versatility of arrangement and freedom of movement between shelter components.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Stock number.
(c) When a first article is required (see 3.2, 4.3 and 6.3).
(d) When an initial production testing sample is required (see 3.3, 4.4 and 6.4).
(e) When an inspection comparison testing sample is required (see 3.4, 4.5 and 6.5).
(f) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(g) When repair kits are required (see 3.10).

6.3 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of Armed Services Regulations ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

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6.5 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

6.6 The following anchoring components are required for the erection of the passageway:

8 each - Bow Spike Assembly conforming to Army Natick Research and Development Command, Drawing 5-4-1570, P/N 5-4-1570-1

16 each - Anchors, ground, arrowhead conforming to type II, grade B of MIL-A-3962 (with one driving rod, driving head and holding handle)

6.7 Cementing of coated cloth. The following system has been used in preparing seams where chlorosulphonated polyethylene coated surfaces of coated cloths are to be adhered to chloroprene coated surfaces:

a. Abrade, by hand, the chlorosulphonated polyethylene surface only to the point of dullness and uniform roughness. Norton Company's Lightning Metalite Cloth - Aluminum Oxide 80-J or 3-M Company's Three-M-ite-Elek-Tro-Cut Cloth J-wt Aluminum Oxide are satisfactory for use in abrading.

b. Wash the abraded chlorosulphonated polyethylene surface and the nonabraded chloroprene surface with clean cotton cloth swabs wet with toluene.

c. Allow each washed surface to dry at room temperature for 15-20 minutes.

d. Brush apply one coat of adhesive (see 3.5.5) to each washed surface.

e. Allow the adhesive to dry at room temperature for 15-20 minutes.

f. Brush apply a second coat of adhesive.

g. After 15-20 minutes drying, reactivate each adhesive with toluene.

h. Assemble immediately, smooth by hand, followed by the use of a steel roller.

i. Cure the cemented assembly a minimum of 24 hours before movement and a minimum of seven days before straining the cemented joint.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

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* 6.9 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

* 6.10 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modification, correction deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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